
“ ” “ ”

“ ”

- Han--Li--Sun Ψ_p
- Corollary 4.2 ε -
-

X_*

$$r_* > 0$$

$$\Psi_p(X_*,T;T-r_*^2) < 1 + \varepsilon_{\mathrm{HLS}}. \tag{0.1}$$

“ HLS-small ”

1.

Han--Li--Sun β -

$$\partial_t F = \frac{\cos^2 \alpha \, H - \beta \sin^2 \alpha \, V}{\cos^2 \alpha + \beta \sin^2 \alpha}, \qquad \beta \geq 0, \tag{1.1}$$

$$F:\Sigma\times[0,T)\rightarrow M^4-\Sigma \qquad M \qquad \text{K\"ahler}$$

$$\cos \alpha = *F_t^* \omega \geq \delta > 0. \tag{1.2}$$

$$Q(t):=\max_{x\in\Sigma}|A|^2(x,t). \tag{1.3}$$

T

$$\limsup_{t\uparrow T}Q(t)=\infty,\qquad \sup_{0\leq t<T}(T-t)Q(t)<\infty. \tag{1.4}$$

“ ”

$$\mathcal{B}_T := \left\{ X \in M : \begin{array}{l} \mathbb{I} \, t_j \uparrow T \, \mathbb{I} \, x_j \in \Sigma, \, |A|(x_j, t_j) \rightarrow \infty, \\ F(x_j, t_j) \rightarrow X \end{array} \right\}. \tag{1.5}$$

$$\begin{array}{ccccccc} & T & & & Q(t) & & t_j \uparrow T \\ Q(t_j) \rightarrow \infty & & x_j & & M & & F(x_j, t_j) \\ & & & & Q(t) \, \mathbb{I} & \implies & \mathcal{B}_T \neq \emptyset. \end{array} \tag{1.6}$$

2.

$$\begin{array}{ccccccc} p & \text{Han--Li--Sun} & & \text{Corollary 4.2} & & & \\ & & & & & & \\ & & & & \Psi_p(X_0, t_0; t) = \int_{\Sigma_t} \frac{1}{\cos^p \alpha} \phi(F) \rho_{X_0, t_0}(F, t) d\mu_t. & & (2.1) \\ & & & & & & \\ r > 0 & (X_0, T) & & T - r^2 \geq 0 & & \text{Corollary 4.2} & \\ \text{HLS} & & 3 & & 0 < 2r < i_M & & \mathcal{R}(X_0, T) \quad \varepsilon_{\text{HLS}} > 0 \quad \text{Corollary} \\ 4.2 & & p, \delta & & & & \end{array}$$

(W)

$$\mathcal{B}_T \neq \emptyset \qquad \qquad \qquad X_* \in \mathcal{B}_T, \qquad r_* \in \mathcal{R}(X_*, T), \tag{2.2}$$

$$\Psi_p(X_*, T; T - r_*^2) < 1 + \varepsilon_{\text{HLS}}. \tag{2.3}$$

(W)

$$\mathcal{B}_T \neq \emptyset \quad \implies \quad \inf_{X \in \mathcal{B}_T} \inf_{r \in \mathcal{R}(X, T)} \Psi_p(X, T; T - r^2) < 1 + \varepsilon_{\text{HLS}}, \tag{2.4}$$

$$(X_*, r_*) \qquad \qquad \qquad \mathcal{B}_T = \emptyset \quad (\text{W})$$

(1.6)

3.

$$\begin{array}{ccccccc} & 3.1 & & \text{HLS} & & & \\ & & & & & & \\ F & & 1 & & T & & (\text{W}) \\ & & & & & & \\ & & & & \sup_{\Sigma \times [0, T)} |A| < \infty. & & (3.1) \\ & & & & & & \\ T & & & \text{Han--Li--Sun Theorem 3.1} & & \eta > 0 & [0, T + \eta) \end{array}$$

$$\begin{array}{ccccccc} (3.1) & & [0,T'] & T' < T & \Sigma & & t \uparrow T & (1.6) & \mathcal{B}_T \neq \\ \emptyset & (W) & X_* \in \mathcal{B}_T & r_* & (2.3) & X_* \in \mathcal{B}_T & t_j \uparrow T & x_j \in \Sigma \end{array}$$

$$|A|(x_j,t_j)\longrightarrow \infty. \tag{3.2}$$

$$F(x_j,t_j)\longrightarrow X_*.\tag{3.3}$$

$$(W)$$

$$\Psi_p(X_*,T;T-r_*^2)<1+\varepsilon_{\mathrm{HLS}}.\tag{3.4}$$

$$\text{Han--Li--Sun Corollary 4.2}$$

$$P(X_*,T;r_*/2)=B_{r_*/2}(X_*)\times (T-r_*^2/4,T] \tag{3.5}$$

$$|A|\leq \frac{C_0}{r_*}.\tag{3.6}$$

$$(3.3) \quad t_j \uparrow T \qquad j$$

$$(F(x_j,t_j),t_j)\in P(X_*,T;r_*/2).\tag{3.7}$$

$$(3.6)$$

$$|A|^2(x_j,t_j)\leq \frac{C_0^2}{r_*^2},\tag{3.8}$$

$$(3.2) \qquad (3.1) \qquad \text{Theorem 3.1} \qquad T \qquad (1.4)$$

$$3.2$$

$$\begin{array}{ccccc} & Q(t)\leq C/(T-t) & (W) & & \\ (W) & \text{“} & \text{”} & & (W) \end{array}$$

$$4.$$

$$\text{Corollary 4.2} \qquad \text{“} \qquad 1+\varepsilon_{\mathrm{HLS}}\text{”} \qquad (W) \qquad X_*\in \mathcal{B}_T \quad X_*$$

$$\bullet \qquad X\in \mathcal{B}_T$$

-
-
-
- $\Theta_p^w < 2$

$$\text{“ } \Psi_p \text{ HLS Corollary 4.2”} \tag{W}$$

$$\tag{W}$$

5.

$$X$$

$$\begin{aligned} \text{(A)} \quad & \exists r \in \mathcal{R}(X,T) : \Psi_p(X,T;T-r^2) < 1 + \varepsilon_{\text{HLS}}; \\ \text{(B)} \quad & \liminf_{r \downarrow 0} \Psi_p(X,T;T-r^2) < 1 + \varepsilon_{\text{HLS}}; \\ \text{(C)} \quad & \Theta_p^w(X,T) := \lim_{r \downarrow 0} \Psi_p(X,T;T-r^2) < 1 + \varepsilon_{\text{HLS}}. \end{aligned} \tag{5.1}$$

$$\text{(C)} \implies \text{(B)} \implies \text{(A)}. \tag{5.2}$$

$$\text{(A)} \quad \text{Corollary 4.2} \tag{C}$$

$$\Theta_p^w < 2 \not\Rightarrow \Theta_p^w < 1 + \varepsilon_{\text{HLS}}. \tag{5.3}$$

$$\text{K\"ahler}$$

$$(1 + \varepsilon_{\text{HLS}}, 2). \tag{5.4}$$

$$\text{weighted Gaussian density} < 2 \tag{W}$$

6.

$$\begin{array}{llll} & & F : \Sigma \times [0,T) \rightarrow M^4 & \text{K\"ahler} \quad \beta- \\ & \cos \alpha \geq \delta > 0 \quad p \quad \text{HLS} & \mathcal{B}_T \quad (1.5) & \mathcal{B}_T \neq \emptyset \\ X_* \in \mathcal{B}_T & r_* > 0 & & \\ & & \Psi_p(X_*,T;T-r_*^2) < 1 + \varepsilon_{\text{HLS}}. & \end{array}$$

$|A|$
 $[0, T)$
 T
 T

“
 T
(W)
”

[1]
X. Han, J. Li and J. Sun, *Gradient Flow for β -Symplectic Critical Surfaces*, Ann. Inst. H. Poincaré C Anal. Non Linéaire 41 (2024), 1083--1116

Theorem 3.1
Theorem 4.1
Corollary 4.2

[2]
M.-T. Wang, *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds*, J. Differential Geom. 57 (2001), 301--338

Proposition 5.2